

Assignment 5

Optimization Dynamics: SGD vs. Adam vs. AdamW

Strict Protocol Edition · Towards Research-Level Competence

Assignment 5

■ GLOBAL RULES

- Form a hypothesis BEFORE every experiment.
- Explain your reasoning before asking questions.
- After using external help, re-implement from memory.
- Every experiment must include: Hypothesis → Result → Interpretation.

Problem Statement

Hold the architecture fixed and systematically compare three optimizers. Before running any experiment, you must write your predictions. Study the loss surface, gradient-to-weight ratios, and implement learning rate warmup from scratch.

- Use your best architecture from Assignment 4 (with residuals + LayerNorm)
- Task: same multi-step reasoning task from Assignment 4
- Three optimizer runs: SGD (with momentum), Adam, AdamW
- You must write your prediction for which optimizer wins BEFORE running the experiments
- Implement learning rate warmup from scratch and test with/without for each optimizer

Implementation Requirements

1. Implement Adam and AdamW from scratch (no optimizer library for the core logic)
2. Implement cosine LR schedule and linear warmup manually
3. Track gradient-to-weight ratio (grad norm / weight norm) -- flag instability when ratio > 1
4. Use the same random seed and dataset for all optimizer runs
5. Log the second moment estimates (v) in Adam -- watch for exploding adaptive rates

Instrumentation Requirements

1. Plot loss curves for all three optimizers on the same chart
2. Log gradient norms AND weight norms separately (not just their ratio)
3. Track the effective learning rate per-parameter in Adam ($\text{lr} / \sqrt{v} + \text{eps}$)
4. Measure and plot gradient-to-weight ratio across layers over training
5. Finite-difference approximation of loss surface curvature at convergence

Failure Analysis (Required)

You MUST document failures. Describe what failed, why you think it failed, test at least two fixes, and evaluate which fix worked and why.

1. Does SGD diverge without warmup? At what LR does it become unstable?
2. Does AdamW actually outperform Adam on this task? Why or why not?
3. What is the effect of beta1 and beta2 in Adam? Run ablations on these.
4. Does warmup help all optimizers equally, or is it more critical for some?

Mathematical Integration

1. Derive the Adam update rule and explain the bias correction terms
2. Explain why weight decay in AdamW differs from L2 regularization in Adam
3. Compute the loss surface curvature using finite differences and relate it to optimizer step size
4. Analyze the relationship between gradient-to-weight ratio and training instability

Research Questions to Answer

- Why does warmup help -- what does it prevent in the early steps of Adam?
- Is there a regime where SGD outperforms Adam on this task?
- What does the gradient-to-weight ratio tell you about learning dynamics?
- Can you predict convergence speed from the loss surface curvature measurement?

Phase Progression

Phase	Focus
Phase 1	Build the core architecture
Phase 2	Compare with baseline / non-attention models
Phase 3	Study optimization dynamics

Phase 4	Modify loss functions and evaluate
Phase 5	Alter architecture systematically
Phase 6	Design novel training techniques

Document everything like a research paper. Your goal: independently analyze neural architectures, diagnose training failures, and propose novel improvements grounded in theory and empirical evidence.